Throughout my education and career, I've moved across different domains, but consistently toward data-focused work. During my bachelor's, I realised I was more interested in data and statistics than traditional software development, which led me to pursue data science and AI in my master's at Uppsala. My experiences have ranged from building a database for an underfunded school to analysing network data for my thesis. Across these, I've leaned toward R&D-oriented roles, as I'm more motivated by exploring new possibilities than maintaining existing systems. While consultancy seemed like a natural path, the constant focus on delivering acceptable solutions within tight constraints limited the depth of work, which didn't suit how I prefer to approach problems.

Growing up in a hospital compound where my father worked gave me early exposure to healthcare systems. I volunteered there and saw firsthand how difficult it can be to standardise data and terminology, even within a single department. During my master's, I built on this interest through courses in healthcare innovation, covering the legal and business aspects of healthtech in Sweden, as well as a project with Akademiska Sjukhuset and a startup focused on interoperability challenges, EHR systems, and health data standards. These experiences have shaped my understanding of the Swedish healthcare landscape.

What drew me to this position specifically is the framing of the problem. Most cancer AI work I've encountered applies machine learning to omics data as a prediction task and stops there. The premise of building a unified mechanistic model, one that captures how metabolism, signalling, and gene regulation actually interact rather than treating them as separate inputs, is a meaningfully different goal. A friend of mine underwent personalised cancer treatment in Sweden, and watching that process made me more aware of both how far precision medicine has come and how much residual uncertainty still comes from incomplete mechanistic understanding rather than a lack of data. The drug response problem in particular, where the same genetic alteration can behave very differently depending on the metabolic state of the cell, feels like exactly the kind of question that needs a model built from biological constraints rather than one that learns statistical associations from the top down.

On the technical side, my master's thesis has given me solid experience with transformer architectures for classification and fine-tuning, in a domain I chose knowing I'd have limited supervisor support, which meant building domain understanding independently. More directly relevant, I've been working since early this year on a feasibility pilot with the Haematology and Oncology department at King Faisal Specialist Hospital in Riyadh. The aim is an AI layer for interpreting capillary electrophoresis outputs, classifying abnormal patterns in curve data and stratifying cases by confidence. Working through the clinical AI literature to scope it has grounded me in how AI systems need to behave in clinical contexts, not just technically. I also have direct experience with LSTM architectures from a time series forecasting project, along with SHAP analysis for feature importance and interpretability, which connects to what the lab describes as building models that are interpretable rather than purely predictive.

I don't have a formal background in computational biology, bioinformatics, or multi-omics data analysis. That is the honest gap. What I can offer is a technical foundation in Python and deep learning, direct experience working on clinical AI at

the edges of my own domain knowledge, and a genuine interest in research that is mechanistically grounded rather than purely empirical. I'm undecided about whether I'll remain in academia after completion, but the translational and open-science orientation of the DDLS programme keeps both options real. My question for the group would be about the biological constraints: how much the mechanistic priors built into the architecture shape what the model is actually capable of learning, and how you evaluate whether the model has captured something biologically meaningful versus found a predictive shortcut.